

- 1 The enzyme urease catalyses the breakdown of urea into carbonate ions and ammonium ions as shown in Fig. 1.1.

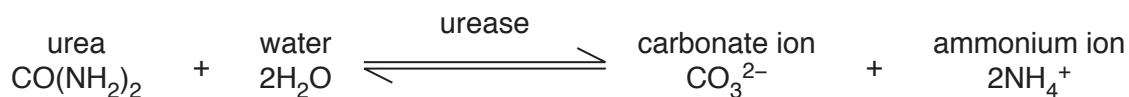


Fig. 1.1

A solution of urea does not conduct electricity, but a solution of the two ions does conduct electricity.

An increase in ions increases conductivity, so the rate of reaction is proportional to the conductivity.

A meter measures conductivity in microsiemens per centimetre ($\mu\text{S cm}^{-1}$). The conductivity meter also records temperature.

K_m shows the affinity of an enzyme for its substrate. The lower the K_m the greater the affinity. A student carried out an investigation to find the K_m of the enzyme urease at different temperatures.

The student:

- made 500 cm^3 of 0.2 mol dm^{-3} solution of urea, molar mass 60.1 g mol^{-1}
- used serial dilution to make a total of five urea solutions from 0.2 mol dm^{-3} stock solution
- used $1\text{ g per } 10\text{ cm}^3$ urease solution
- used a conductivity meter to measure the initial rate of reaction at each temperature.

Fig. 1.2 shows the experimental set-up during the course of a reaction.

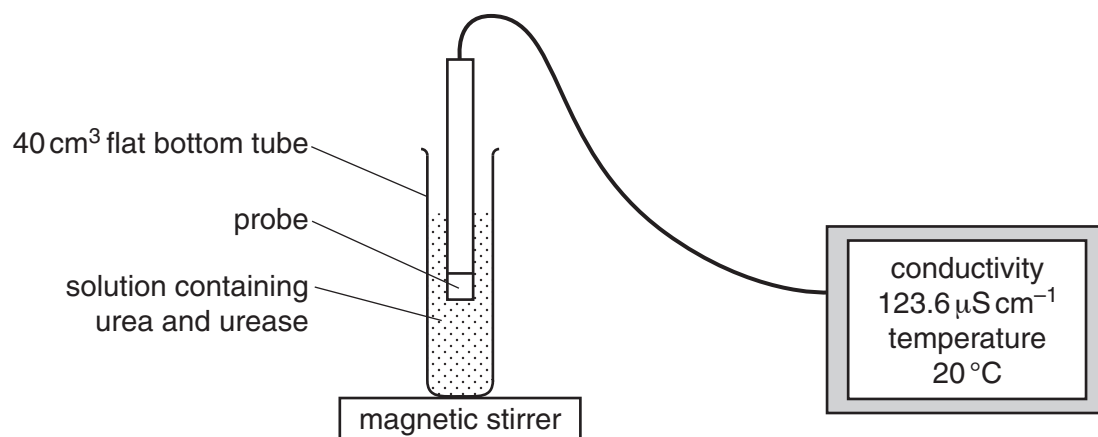


Fig. 1.2

(a) (i) Describe how the student could make 500 cm³ of 0.2 mol dm⁻³ solution of urea.

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(ii) Describe how the student made a further **four** solutions of urea by **serial dilution**.

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(b) (i) Identify the independent and dependent variables in this investigation.

independent

dependent [2]

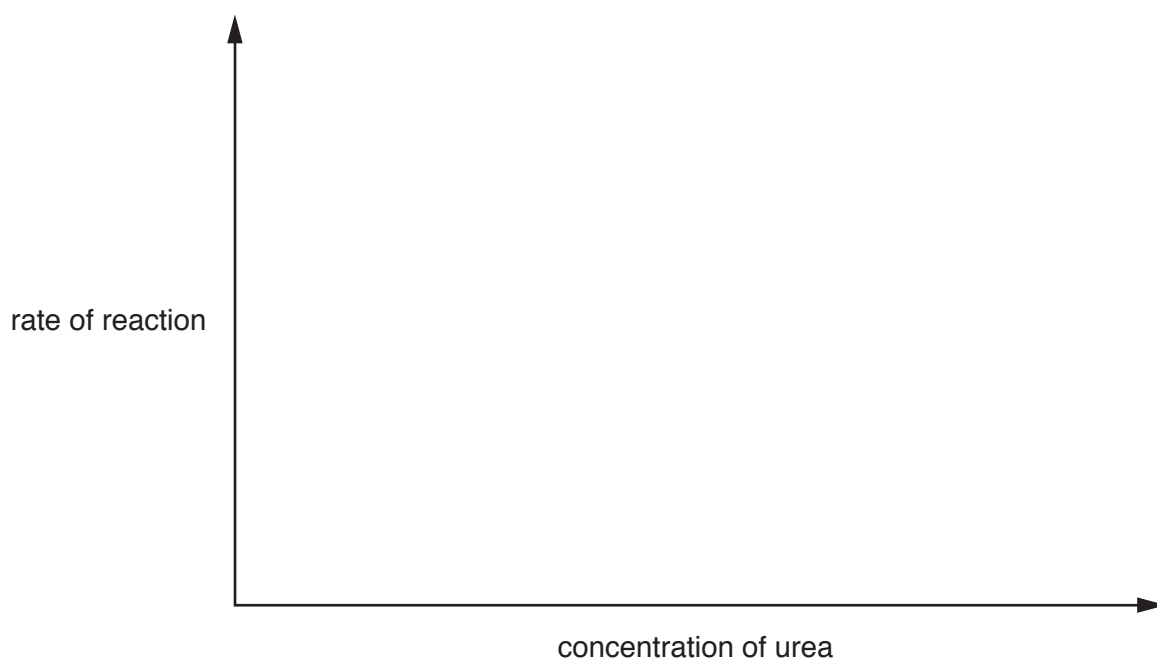
(ii) Suggest a suitable control for this investigation.

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- (c) (i) Use the axes below to sketch a graph to show the effect of substrate concentration on the initial rate of reaction at **one** temperature.

Indicate on your graph how the student could find the K_m of urease at that temperature.



[4]

- (ii) Table 1.1 shows the results that the student obtained for K_m at different temperatures.

Table 1.1

temperature	A	B	C	D	E
K_m /arbitrary units	1.8	1.3	0.9	0.8	1.2

State which of these temperatures, **A** to **E**, is closest to the likely optimum temperature of urease.

Explain your answer.

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[Total: 19]